

UniFlow University ERP & Virtual Labs

Server Infrastructure Sizing & Production Deployment Plan

PREPARED FOR
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RECOMMENDED SETUP
4-Node Production Cluster

DATE & VERSION
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CLASSIFICATION
Confidential Tech Report

1. EXECUTIVE SUMMARY: SERVER COUNT & SEPARATION RATIONALE

To host the **UniFlow Platform** with high-demand extensions (BigBlueButton Virtual Classrooms, Virtual Programming Labs (VPL), Medical/Science Simulators, and AI Proctoring), an enterprise workload-separated topology is mandatory.

Total Required Nodes: Exactly 4 Dedicated Servers + S3 Object Storage

Why separate nodes? Video transcoding (WebRTC UDP), untrusted student code execution (VPL Jails), and financial/student database transactions have conflicting resource patterns. Isolating them prevents exam crashes, guarantees 99.9% uptime, and eliminates security leak vectors.

2. EXACT HARDWARE SPECIFICATIONS & WORKLOAD BREAKDOWN

Server 1: Core Web & Application Gateway

srv-uniflow-app

VCPU	RAM	STORAGE	NETWORK	OS
4 - 8 Cores	16 GB DDR4/5	120 GB NVMe	1 Gbps Public	Ubuntu 24.04 LTS

- UniFlow Core App:** Next.js 16 (React 19, TypeScript) on Node.js 22 LTS under PM2 cluster mode (4-6 workers).
- Reverse Proxy / Edge:** Nginx with HTTP/2, TLS 1.3, Let's Encrypt Certbot auto-renewal, and Brotli/Gzip compression.
- Cache & Task Queue:** Redis in-memory store for session tokens, real-time push notifications, and API rate limiting.
- AI Gateway & Proctoring Controller:** Webhook telemetry ingestion for AI proctoring and LLM tutoring proxies.

Server 2: Dedicated Database & Isolation Tier

srv-uniflow-db

VCPU	RAM	STORAGE	NETWORK	OS
4 - 8 Cores	16 - 32 GB	300 GB Enterprise NVMe	1 Gbps Private LAN	Ubuntu 24.04 LTS

- PostgreSQL 17 Engine:** Tuned buffer cache (shared_buffers=8GB, work_mem=64MB, effective_cache_size=24GB).
- PgBouncer Connection Pooler:** Port 6543 in Transaction Mode for restricted uniflow_app runtime role.
- Dual-Role RLS Security:** Strict Row-Level Security isolation; direct port 5432 reserved for owner schema migrations.
- Automated Backups & Archiving:** WAL-G / pgBackRest for continuous Point-in-Time Recovery (PITR) replicated to S3.

Server 3: BigBlueButton (BBB) Video & Virtual Classrooms

srv-uniflow-bbb

VCPU	RAM	STORAGE	NETWORK	OS
8 - 16 Cores	16 - 32 GB	300 GB NVMe (Buffer)	1 Gbps Unmetered	Ubuntu 22.04 LTS

- **BigBlueButton 2.7+ Suite:** FreeSWITCH (SIP/VoIP audio), Kurento/Mediasoup (WebRTC video & screen sharing), BBB HTML5 client.
- **Coturn STUN/TURN Server:** Traversal server allowing students behind restricted firewalls/cellular NATs to stream smoothly.
- **Recording Processing Pipeline:** Converts raw audio/video to web formats and pushes finished recordings directly to S3.
- **Internal BBB Nginx & Redis:** Manages WebSockets and live meeting state synchronization without touching core ERP.

Server 4: VPL Virtual Programming Labs & Simulators

srv-uniflow-vpl

VCPU	RAM	STORAGE	NETWORK	OS
4 - 8 Cores	8 - 16 GB	80 GB NVMe	1 Gbps Private / Firewallled	Ubuntu 24.04 LTS

- **vpl-jail-system Sandbox Daemon:** Sandboxed execution engine with Linux cgroups (Max 5s CPU time, 256MB RAM per run).
- **Compilers & Language Toolchains:** GCC, G++, Python 3, OpenJDK (Java 17/21), Node.js, Rust, Go, and PHP.
- **Medical & Science Simulators Hub:** Docker Engine hosting WebAssembly/WebGL 3D simulation modules with LTI 1.3 sync.

Auxiliary Cloud Tier: S3-Compatible Object Storage

AWS S3 / Wasabi / MinIO

CAPACITY	DURABILITY	TYPE
1 TB - 5 TB (Scalable)	99.99999999% (11 9s)	Hot Object Storage

Stores BBB video recordings, student identity/medical document attachments, and daily encrypted database disaster-recovery snapshots.

3. SERVER SPECS & NETWORK MATRIX AT A GLANCE

SERVER ROLE	HOSTNAME	VCPU	RAM	DISK (NVME)	INBOUND PORTS	NETWORK EXPOSURE
Core Web & Gateway	srv-uniflow-app	4 - 8	16 GB	120 GB	80, 443 (HTTP/S)	PUBLIC EDGE
Database & PgBouncer	srv-uniflow-db	4 - 8	16 - 32 GB	300 GB	6543 (App), 5432 (Owner)	PRIVATE LAN
BigBlueButton (BBB)	srv-uniflow-bbb	8 - 16	16 - 32 GB	300 GB	80, 443, UDP 16384-32768	PUBLIC MEDIA
VPL Jail & Simulators	srv-uniflow-vpl	4 - 8	8 - 16 GB	80 GB	8080 (VPL Internal)	FIREWALL FILTERED

Budget 3-Server Alternative: Small universities (<500 active students) can combine **Server 1** and **Server 2** into one high-spec node (8 vCPU, 32 GB RAM). **Server 3 (BBB)** and **Server 4 (VPL)** must always remain dedicated for WebRTC stability and sandbox security.

4. PHASED PRODUCTION DEPLOYMENT ROADMAP

Phase 1: Base Hardening & VPC

Provision Ubuntu nodes; configure SSH Ed25519 keys, UFW firewalls, Fail2ban brute-force defense, and private LAN network.

Phase 2: Database & PgBouncer

Install PostgreSQL 17 + PgBouncer on DB node. Setup dual roles (uniflow_app vs uniflow) and WAL-G automated S3 archiving.

Phase 3: UniFlow App Deployment

Deploy Next.js 16 standalone build under PM2. Run migrations and execute `npm run db:check-roles` to guarantee tenant RLS isolation.

Phase 4: BigBlueButton & Coturn

Deploy BBB 2.7+ with FreeSWITCH, Kurento/Mediasoup, Coturn STUN/TURN, and post-recording S3 auto-export scripts.

Phase 5: VPL Sandbox & Simulators

Install `vp1-jail-system` with cgroups CPU/memory quotas. Connect LTI 1.3 Advantage authentication for medical/science labs.

Phase 6: Edge Proxy & Monitoring

Deploy Nginx with TLS 1.3 + Cloudflare WAF. Configure Prometheus, Grafana, and Uptime Kuma for 24/7 alerting and health checks.

5. MONTHLY HOSTING COST ESTIMATES

INFRASTRUCTURE COMPONENT	CLOUD VPS / BARE-METAL (HETZNER / OVH / DO)	HYPERSCALER (AWS / AZURE MANAGED)
Server 1: Web App & Redis Gateway	\$40 – \$90 / mo	\$160 – \$380 / mo
Server 2: PostgreSQL 17 + PgBouncer	\$60 – \$140 / mo	\$220 – \$550 / mo
Server 3: BigBlueButton Video Node	\$80 – \$220 / mo	\$300 – \$750 / mo
Server 4: VPL Sandbox & Simulators	\$40 – \$100 / mo	\$140 – \$320 / mo
S3 Storage (1–5 TB) & Offsite Backups	\$15 – \$45 / mo (Wasabi / MinIO)	\$50 – \$160 / mo (AWS S3)
WAF / CDN / Monitoring / SSL	\$20 – \$50 / mo (Cloudflare Pro)	\$80 – \$200 / mo
Estimated Monthly Total	\$255 – \$645 / month	\$950 – \$2,560 / month
Estimated Annual Total	\$3,000 – \$7,700 / year	\$11,400 – \$30,700 / year

6. MANDATORY SECURITY & PRODUCTION INVARIANTS

Key Architecture Guarantees:

- 1. **Database RLS:** The application connects only via `uniflow_app` behind PgBouncer so Row-Level Security cannot be bypassed.
- 2. **Jail Sandboxing:** VPL executions operate under strictly isolated Linux cgroups with a 5s execution timeout and disabled networking.
- 3. **Disaster Recovery:** Point-in-time database WAL archiving + encrypted daily offsite S3 sync ensures zero data loss.